

National University of Computer Emerging Sciences

Quiz: Operating Systems (Solution)

Name: _____

Roll No: _____

Section: _____

Multiple Choice Questions (1 mark each)

[Total: 5 Marks]

1. What is the main reason for using processes in an operating system? [1]
 - (a) To make program execution faster
 - (b) **To enable concurrency and better system utilization**
 - (c) To eliminate the need for memory management
 - (d) To avoid the need for an operating system

2. Which of the following does not belong to a process's address space? [1]
 - (a) Code segment
 - (b) Data segment
 - (c) CPU registers
 - (d) **Printer buffer**

3. What happens during a context switch? [1]
 - (a) The CPU switches from user mode to kernel mode
 - (b) **The current process state is saved, and a new process state is loaded**
 - (c) The operating system resets the CPU
 - (d) The system enters a deadlock state

4. In the Two-State Process Model, when a process moves from Not Running to Running, what event occurs? [1]
 - (a) Context switch
 - (b) Process creation
 - (c) **Dispatch**

(d) Process termination

5. Which component of the operating system is responsible for switching the CPU from one process to another? [1]

- (a) Dispatcher
- (b) Scheduler
- (c) Process Manager
- (d) Interrupt Handler

Conceptual Questions

[Total: 10 Marks]

1. Differentiate between a CPU-bound and an I/O-bound process. How does their execution pattern impact CPU scheduling? [3]

A **CPU-bound process** performs **intensive computations** and spends most of its time executing instructions, with minimal waiting for I/O operations. In contrast, an **I/O-bound process** primarily interacts with **I/O devices** and spends a significant portion of its time waiting for input/output completion.

Impact on CPU Scheduling: If a system has mostly **CPU-bound processes**, the CPU remains busy while I/O devices are underutilized. Conversely, if most processes are **I/O-bound**, the CPU may remain idle while waiting for I/O operations to complete. A well-balanced mix of **CPU-bound** and **I/O-bound processes** ensures **efficient resource utilization**.

2. How does the dispatcher determine the next process to execute? What role does the short-term scheduler play in this decision? [3]

The **dispatcher** is responsible for switching the CPU from one process to another by performing a **context switch**. It **saves the state of the currently running process** and **loads the state of the next scheduled process**.

The **short-term scheduler** decides which process gets CPU time next. It selects a process from the **ready queue** based on the **scheduling algorithm** (e.g., **FCFS**, **Round Robin**, or **Priority Scheduling**). The dispatcher then executes this decision by performing the actual **process switch**.

3. PCB Structure & Memory Management:

[4]

The Process Control Block (PCB) contains critical information about a process, including its memory limits and address space.

(a) Explain how the memory limit fields in the PCB prevent a process from accessing unauthorized memory regions.

[2]

The **memory limit fields** in the PCB define the **base** and **limit addresses** assigned to a process. If a process attempts to access memory outside this range, the OS triggers a **segmentation fault**, preventing unauthorized access. This ensures **memory protection** and prevents **processes from interfering with each other**.

(b) How does this relate to base and limit registers in modern operating systems?

[2]

The **base register** stores the starting memory address of a process, while the **limit register** defines the upper bound of accessible memory. The OS uses these registers to **validate every memory access**, ensuring that a process does not access unauthorized areas. This technique provides **hardware-enforced memory protection** in modern operating systems.